

# Abhay Verma

## Summary

Database developer with focused experience in relational database management systems. Worked with MySQL for structured data querying, schema design, and data manipulation. Familiar with NoSQL concepts through MongoDB for document-based storage. Comfortable writing complex queries, handling edge cases, and working across both relational and non-relational systems depending on project needs.

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## Tech Skills

| Category             | Details                                                          |
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| Relational Databases | MySQL, PostgreSQL                                                |
| NoSQL Databases      | MongoDB                                                          |
| Query Writing        | Joins, Subqueries, Window Functions, Aggregations, CTEs          |
| Data Operations      | INSERT, UPDATE, DELETE, conditional transforms, string functions |
| Schema Concepts      | Normalization, Foreign Keys, Indexing, NULL handling             |
| Tools                | MySQL Workbench, MongoDB Compass, VS Code                        |

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## Project Experience

**Employee & Department Database — MySQL** Designed and queried a normalized multi-table schema covering employees, departments, salaries, and managers. Wrote self-joins to compare employee records against their managers, used window functions (`DENSE_RANK`) for top-N salary filtering per department, and handled NULL-safe joins for optional relationships like bonuses and unique identifiers.

**Sales & Transaction Database — MySQL** Built queries across sales, orders, and customer tables. Identified customers who visited without transacting using `LEFT JOIN` and NULL filtering. Calculated average selling prices across overlapping date ranges using conditional aggregation and date comparison logic.

**User Activity & Content Database — MySQL** Worked with tweet, article view, and follower tables. Filtered invalid records using string length functions, identified authors who viewed their own articles, and ranked follower counts with `ORDER BY` and `GROUP BY` combinations.

**Document Store Practice — MongoDB** Stored and retrieved flexible records where schema consistency was not required. Practiced queries using filter operators, projection, and working with nested fields and array-based data.

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## Practicing Problems

Solved 25+ SQL problems on LeetCode covering Easy to Hard difficulty.

| Difficulty | Problems |
|------------|----------|
|------------|----------|

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|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Easy | Combine Two Tables, Duplicate Emails, Employees Earning More Than Managers, Rising Temperature, Employee Bonus, Find Customer Referee, Big Countries, Classes With 5+ Students, Sales Person, Swap Sex of Employees, Product Sales Analysis I, Article Views I, Replace Employee ID, Customer Who Visited Without Transaction, Fix Names, Invalid Tweets, Find Followers Count, Recyclable and Low Fat Products |
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|--------|------------------------------------------------------------------|
| Medium | Second Highest Salary, Nth Highest Salary, Average Selling Price |
|--------|------------------------------------------------------------------|

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|------|-------------------------------|
| Hard | Department Top Three Salaries |
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